COMPUTER ORGANIZATION AND ARCHITECTURE
ENCT 303

Year/Part: III/I

Teaching Schedule				Examination Scheme						Total
				Theory			Practical			
					Final			Final		
L	T	P	Total	Assessment Marks	Duration (Hrs)	Marks	Assessment Marks	Duration (Hrs.)	Marks	
3	1	1.5	5.5	40	3	60	25	0	0	125

Depth Codes

E-Explanation	C-Circuit	D-Definition	DM-Demonstration
DV-Derivation	DW-Drawing	P-Proof	I-Illustration
NUM-Numerical	PRG-Programming	S-State	ACT-Activity-based Learning
MP- Mini Project	EXP-Experiment	REV-Review / Recap	PS- Problem Solving
QA- Question Answer	Q- Quiz	ST- Surprise Test	MT-Mid Term Test

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
				L	T	P	
1	**Introduction**						
	1.1 Organization and Architecture	E, D	Computer Organization and Computer Architecture introduction with its differences and significance	0.5			**1**
	1.2 Computer Structure	E, I	Single processor and multicore processors	1			**1**
	1.3 Performance Assessment	NUM, PS	Instruction execution rate: CPI, MIPS, MFLOPS, arithmetic mean, harmonic mean, speed metric, geometric mean, rate metric, Amdahl's law, speed up formulas with numerical examples	1			**1**
	1.4 Instruction Fetch and execute	DW, I	Instruction Fetch and execute cycle state diagram with and without interrupt	1.5			**1**
	Numerical examples on performance assessment	NUM	Numerical		2		**2**
	1.5 Computer component, interconnection structure, bus interconnection, PCI	DW, E	Computer components with inter connection structure of each components. Bus types & PCI interface	0.5			**2**
	1.6 RISC architecture, Overlapped register window Berkeley RISC	E	RISC concepts, overlapped register window and Berkeley RISC explanation	1.5			**2**
	Evaluation	QA, Q					

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan L	T	P	Week
2	**Central Processing Unit (CPU)**						
	2.1 Processor Bus Organization	E	CPU bus signals and flow with diagrams and explanation	1			**3**
	2.2 Processor register Organization: Control word, examples of microoperations	I, E	Control word and microoperations of Control Unit. Various microoperations of instructions	1			**3**
	2.3 Stack Organization Register stack, memory stack, reverse polish notation, evaluation of arithmetic expressions	NUM	Usage of Register and memory stack	1			**3**
	2.4 Instruction formats: CPU organization, zero and more address instruction formats	E, NUM	Evaluation of an expression in 0, 1, 2, 3 address instruction formats	1			**3**
	2.5 Addressing modes: Types, examples, strengths and weaknesses	E, I	Types of Addressing modes with examples. Pros and cons of addressing modes types	1			**4**
	2.6 Instruction set	E	Data transfer instruction, Data manipulation instruction: Arithmetic, logical and shift operations, Program control instruction explanation and examples	1			**4**
	2.7 Status bit conditions	I	Status bit flags Carry, Overflow, Sign, Zero conditions	0.5			**4**
	2.8 Interrupt: Definition, types, processing and ISR	E	Types of Interrupts and ISR flow diagram	0.5			**4**
	Coding examples covering different instruction formats	NUM	Numerical		2		**4,5**
	Evaluation	QA, Q					

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan L	T	P	Week
3	**Control Unit (CU)**						
	3.1 Hardwired control unit	E	Hardwired control unit diagram and explanation	1			**5**
	3.2 Microprogrammed control unit	E, I	Microprogrammed control unit diagram and explanation	1			**5**
	3.3 Microinstructions, control memory organization, Wilkes control	DW	Micro instruction format, control memory, Wilkes control architecture diagram and explanation	1			**5**
	3.4 Microinstruction sequencing: Design considerations, sequencing techniques, address generation, microinstruction encoding	E	Next address generation using micro instruction sequencing	1			**6**
	3.5 Application of microprogramming	E	Areas of microprogramming	0.5			**6**
	3.6 Microinstruction execution	E	Process of micro instruction execution	0.5			**6**
	Microprogramming examples in CU	NUM	Numerical		2		**6**
	Evaluation	QA, Q					

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan L	T	P	Week
4	**Memory System**						
	4.1 Characteristics of memory system	D, E	Characteristics like Access time, cycle time, bandwidth explanation	1			**7**
	4.2 Memory classification and hierarchy	E, DW	Cache memory, Main memory, Virtual memory structures	1			**7**
	4.3 Semiconductor memory and its types, read only memory, read/write memory	E, I	RAM, ROM, Flash memory	1			**7**
	4.4 RAM modules and interfaces: DDR, DIMM and SODIMM	I, E	DDR, DIMM, SODIMM explanation with examples	1			**7**
	4.5 Cache memory	NUM, DW	Cache principles explanation, Elements of cache design: Cache size, mapping function, replacement algorithms, write policy, block size, single and multilevel caches and unified versus split cache Mapping, replacement, write policy	2			**8**
	4.6 External Memory	E	Magnetic Disk, RAID level 1 to 5, optical	1			**8**

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan L	T	P	Week
			memory, Magnetic tape, SSD				
	Cache memory mapping: Hit and miss ratio	NUM	Numerical		2		**8,9**
	Evaluation	QA, Q					

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan L	T	P	Week
5	**Computer Arithmetic**						
	5.1 ALU (Arithmetic and logic unit)	D, E	ALU components and operation, Design steps	1			**9**
	5.2 Integer representation: Sign-magnitude representation, two's complement representation, converting between different bit lengths, fixed-point representation	E, DW	Formats for representation of integers, 2's complement method and conversions	1			**9**
	5.3 Integer arithmetic	E, I	Addition, Subtraction, Multiplication (Unsigned, Signed Booth) and Division algorithm (Restoring and Non Restoring) and flowchart with examples.	3			**9,10**
	5.4 Floating-point arithmetic	I, E	Floating-point representation: Principles, IEEE standard for binary floating-point arithmetic algorithm, Addition, Subtraction, Multiplication and Division algorithm	2			**10**
	Design of arithmetic circuit, logic circuit and ALU	NUM	Numerical		2		**11**
	Numerical examples for various arithmetic algorithms	NUM	Numerical		3		**11,12**
	Evaluation	QA, Q					

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
				L	T	P	
6	**Pipelining and Vector Processing**						
	6.1 Pipelining and its importance	E	Different stages of Pipelining and their significances	0.5			**12**
	6.2 Instruction and arithmetic pipelining	E	Pipelining structures	0.5			**12**
	6.3 Pipelining hazards: Data, structural and control hazards	PS	Data hazards, Structural Hazards, Control Hazards	1			**12**
	6.4 RISC pipeline	I	RISC execution stages	0.5			**12**
	6.5 Parallel processing	E	Parallel Processing types	0.5			**12**
	6.6 Vector processing	E	Vector operations, Matrix multiplication, Memory interleaving, Superscalar processors, Supercomputers explanation	0.5			**13**
	6.7 Array processors: Attached array processor and SIMD array processor	E	Explanation of Array processor types	0.5			**13**
	Evaluation	QA, Q					

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
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7	**Input/ Output**						
	7.1 External devices	E	External device characteristics	0.5			**13**
	7.2 I/O modules: Module function, module structure	E	Block diagram of I/O modules with functions	0.5	0.5		**13**
	7.3 Programmed I/O, I/O commands, I/O instructions, flowchart	PS	Flowcharts with explanation	0.5	0.5		**13**
	7.4 Interrupt driven I/O, interrupt processing and flowchart	I	ISR, flowcharts	0.5	0.5		**13**
	7.5 Direct memory access (DMA): Drawbacks of programmed and interrupt	E	Comparison, DMA controller block diagram, working principles	1	0.5		**13,14**

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan L	T	P	Week
	driven I/O, DMA function, typical DMA block diagram and possible DMA configuration						
	7.6 I/O channels and processors: The evolution of the I/O function, characteristics of I/O channels	E	I/O channels and processors, features of I/O channels	1			**14**
	7.7 The external interface: Types of interfaces, point-to point and multiple configurations, small computer system interface (SCSI)	E	Interface types, Point to point and multiple configurations, SCSI	1			**14**
	Evaluation	QA, Q					

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan L	T	P	Week
8	**Multiprocessor System**						
	8.1 Multiprocessor computers and their characteristics	E	Multiprocessor computers definition and characteristics	1			**15**
	8.2 Multi-core computers and their architecture	E, I	Multicore architecture	0.5			**15**
	8.3 Interconnection structure: Time-shared common bus, multiport memory, crossbar switch, multistage switching network and hypercube system	DW	Explanation of types of interconnection structures with diagram. Comparison between them	1.5			**15**
	8.4 Interprocessor arbitration	E	Explanation of Arbitation	1			**15**
	8.5 Interprocessor communication and Synchronization	DW	Explanation with synchronizing process	1			**15**
	Evaluation	QA, Q					

References: (Primarily based on the syllabus, and relevant chapters may be consulted as needed)

1. Stallings, W. (2018). Computer organization and architecture. Prentice Hall of India.
2. Mano, M. M. (2008). Computer system architecture. Pearson Education.
3. Hennessy, J. L., Patterson, D. A. (2000). Computer architecture: A quantitative approach. Harcourt Asia.